Finding Needles in Document Haystacks: Augmenting Serendipitous Claim Retrieval Workflows

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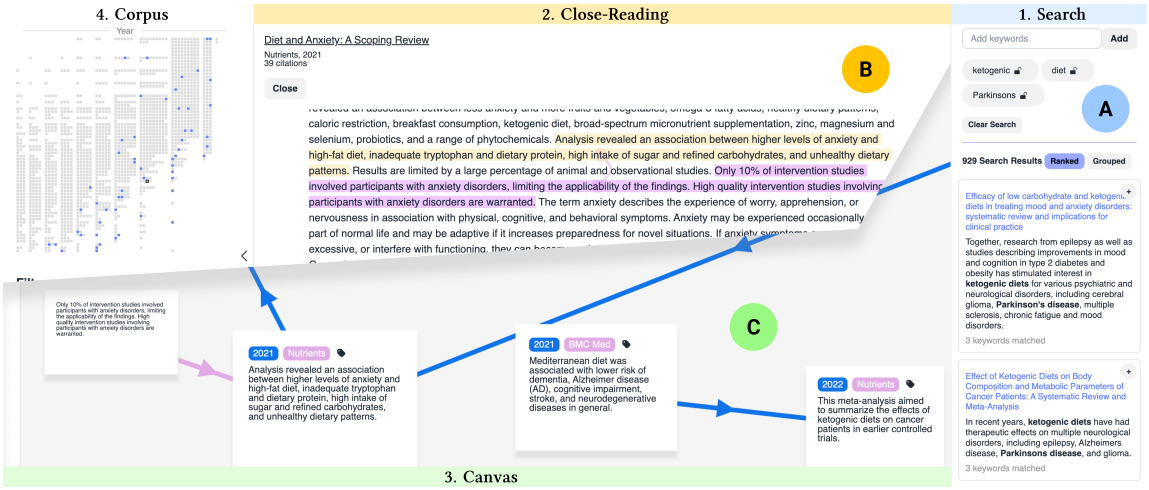


Figure 1: The *NEEDLE* interface with 1) *Search Panel* to support keyword-based retrieval of individual claims, 2) *Close-Reading View* to contextualize findings, 3) *Canvas* to consolidate results, and 4) *Corpus View* to provide an interactive overview of documents and search results. The tool instantiates our claim-based workflow using natural language inference, supporting users in the efficient preliminary exploration of large text corpora.

Abstract

Preliminary exploration of vast text corpora for generating and validating hypotheses, typical in academic inquiry, requires flexible navigation and rapid validation of claims. Navigating the corpus by titles, summaries, and abstracts might neglect information, whereas identifying the relevant context-specific claims through in-depth reading is unfeasible with rapidly increasing publication numbers. Our paper identifies three typical user pathways for hypothesis exploration and operationalizes sentence-based retrieval combined with effective contextualization and provenance tracking in a unified workflow. We contribute an interface that augments the previously laborious tasks of claim identification and consistency checking using NLP techniques while balancing user control and serendipity. Use cases, expert interviews, and a user study with 10 participants demonstrate how the proposed workflow enables users to traverse literature corpora in novel and efficient ways. For the evaluation, we instantiate the tool within two independent domains, providing novel insights into the analysis of political discourse and medical research.

CCS Concepts

• **Human-centered computing** → **Web-based interaction**; **User studies**.

Keywords

human-AI interaction, natural language processing, provenance, serendipity, text data

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1 Introduction

Discovering valuable insights from the vast and rapidly growing pool of accessible information can be as challenging as finding a needle in a haystack. This problem is particularly pertinent to the research community, where the number of published papers keeps increasing, making it difficult to identify relevant claims in the text using manual approaches. This information overload exists in most major application domains. For example, the biomedical field alone is reporting upwards of a million new published papers every year [17], while in the AI domain, new authors are entering the field every hour [42].

Thus far, the attempts to help users navigate these overwhelming text corpora have relied on disregarding individual claims in favor of more high-level solutions such as curating content, creating summaries, and improving productivity [14, 27]. The widespread adoption of large language models (LLMs) has automated many of these summarization methods and heightened the focus on higher level understanding [30].

These approaches, however, have several limitations, as abstracting away from individual claims and only navigating the text corpora using a selection of titles, abstracts, and summaries (i.e., existing information hierarchies) can lead to premature conclusions, lack of specificity, and missed details [22, 26]. Losing access to specific statements also erodes user agency as auto-generated summaries and algorithms inherently decide which information to prioritize [6]. On the other hand, a claim-based approach (i.e. focusing on directly quoted sentences from main text bodies) can help users retain their capacity to engage with nuances and subtleties in the text [3]. This is especially important in the early stages of information discovery, where the user is performing a preliminary exploration to construct an initial hypothesis regarding a topic. This stage often necessitates serendipitous (i.e. unexpected yet exciting) new connections and rapidly parsing a plethora of tangentially related ideas that might transcend disciplines [44]. While the existing solutions to aid hypothesis exploration are insufficient, expecting users to simply conduct close reading of extensive texts without support is equally impractical.

In this paper, we contribute a novel workflow to facilitate efficient hypothesis generation and validation. At the highest level, the solution unifies three typical user pathways and adopts a serendipitous claim-based approach to navigate large text corpora without relying on existing information hierarchies. More specifically, by tightly integrating claim search, contextualization, and provenance tracking functionalities, we reinforce user agency and support rapid validation and consistency checking of hypotheses. Our workflow caters primarily to experts with knowledge in the domain who, in the preliminary stages of research, are looking to obtain an overview, explore ideas, and link concepts.

We instantiate the workflow with *NEEDLE* , an interactive interface, and test it in the context of two independent domains: medical research and political discourse. Augmenting features like claim retrieval and consistency checking using natural language processing (NLP) techniques provides experts with the tools to efficiently validate hypotheses. Simultaneously, various integrated components support the users in the search, organization, and consolidation of claims. The analysis is driven by a user-centric and serendipitous approach, which is key to discovering unique insights.

Our solution is evaluated through a user study with ten research scientists and interviews with two experts from the medical domain. Additionally, a use case from the political domain is used to further highlight the capabilities and limitations of the tool. We conclude the paper with a discussion regarding directions for future research. Overall, our main contributions are as follows:

(1) **Claim-based workflow** - a unified approach combining sentence-based retrieval with efficient contextualization and provenance tracking for three typical user pathways;
(2) **Interactive interface** - a design that enhances claim identification and consistency checking while maintaining a balance between user control and serendipity;
(3) **Systematic evaluation** - a comprehensive evaluation involving a use case, expert studies, and user studies to verify the effectiveness of the system.

2 Related Work

We review related work focusing on two key aspects of preliminary literature exploration: literature analysis tools and user workflows for scientific literature exploration. In the following, we demonstrate that while various methods exist to address specific aspects of preliminary literature exploration, none of the current solutions fully support the workflow proposed in this paper.

(Scientific) Literature Analysis. Literature research is a foundational component of modern scientific inquiry, enabling researchers to synthesize existing knowledge, identify gaps, and guide future studies [40, 49]. Traditional bibliographic databases like *Google Scholar*, *Web of Science*, and *Microsoft Academic* are widely used across disciplines for retrieving information and navigating research documents [32]. Recent advancements leverage LLMs for information retrieval purposes [53]. This includes tools such as *Elicit*[1] and *Consensus*[2] to generate summaries and facilitate research tasks, as well as fully automated research agents that attempt to streamline the scientific paper-writing workflow [30]. These automated approaches, although effective in some cases, are accompanied by the well-documented inherent limitations of LLMs themselves. Notably, they often struggle to ensure the accuracy and reliability of generated content, potentially leading to misinformation [21].

Various interactive analytics tools aim to facilitate more in-depth engagement with these large text corpora. For instance, visual literature browsers employ variations of faceted search and node-link representations to explore authors and publications [12] and enable users to filter and browse the literature collection utilizing textual search, keyword, and meta-information filters [5]. In these tools, visualizations are used to showcase keyword frequencies and temporal contexts, as well as to examine relationships among selected items [5]. *Bridger* [37] looks to solve the problem of information

[1] https://elicit.com/
[2] https://consensus.app/

overload by introducing a system that facilitates the discovery of authors and their work. By leveraging multidimensional author representations, users can uncover novel and potentially useful research directions. In addition to citation and meta-data information, some tools enable literature exploration [45] and cross-domain research [23] based on topic model results. Topic models are particularly used to provide a structured overview of research landscapes [4, 51]; however, they often lack the nuance required for in-depth analysis [8].

Recently, diverse literature analysis tools have emerged that utilize LLMs for information search and personalization purposes. For instance, DiscipLink is an interactive system tailored to interdisciplinary information seeking, featuring a human-AI co-exploration workflow in which an LLM suggests exploratory questions for investigation [52]. *Selenite* [28] employs LLMs as reasoning machines and knowledge retrievers to automatically generate a comprehensive overview of options, serving as a starting point for the user's sensemaking process. Similarly, *Sensescape* [41] enables users to explore topics by querying LLMs and organizing results into a hierarchical visual structure that facilitates navigation, tracks provenance, and suggests further avenues for exploration. Like topic model approaches, these tools focus on high-level thematic exploration without addressing the nuances embedded in specific claims, and come along with challenges such as lack of explainability and transparency of LLM decisions [50]. *TeleoScope* [7] offers a visual editor for exploratory thematic analysis, allowing users to refine document similarity metrics to identify relevant themes. The document similarity is captured by an LLM learning from user interactions. A workspace that incorporates a more fine-grained analysis is presented in the work *Marco* [15], which integrates LLM-based assistants that capture user information needs through natural language inputs and generate summaries, tailored for business workflows. Although the last two tools exhibit some similarities to the *NEEDLE* interface, showcased in this paper, there are some significant differences. The *Telescope* interface is designed for short documents (e.g., social media posts) and does not address navigating within a document (e.g., exploring a set of claims), nor the associated interaction dynamics. *Marco*, however, supports structured artifacts like tables of extracted information, rather than preliminary exploration artifacts such as mind maps or unstructured notes.

There are also specialized tools, which concentrate only on a specific aspect of a more complex literature analysis workflow. For instance, *Passages* [18] presents an abstract framework for the interconnection of text information across multiple applications while maintaining the provenance of information from different sources. *Threddy* [25] connects a selected text snippet to its referenced papers and links them to the citation context. *ScholarPhi* [29], an enhanced reading interface, offers explanatory features like tooltips that display definitions from a paper alongside its content and highlight how terms and symbols are used throughout the paper. Lastly, the *Semantic Reader Project* [19] explores the automatic creation of dynamic reading interfaces, supporting standard reading features (e.g., table of contents), and meaningful augmentations. Other tools explore the power of serendipity in information seeking, research, and ideation, which is a relevant concept for this work, while utilizing the serendipity concept only for a high-level topic model analysis [44] or analysis of specific content- or metadata aspects (e.g., keyword chains) [2], rather than claims. Although each of these tools supports a specific task, context switching and managing multiple separate tools can lead to a number of challenges [24], making the integration and seamless transitions between these isolated components just as important as the components themselves. This is why a contribution that enhances workflow integration is valuable.

Workflows for (Scientific) Literature Analysis. In the past, researchers have introduced multiple frameworks that outline user workflows for information search and knowledge generation scenarios. For instance, the *Information Foraging Theory* by Pirolli and Card [35] aims to understand how humans search for information and how technologies for information gathering are adapted. The *Sensemaking Process* [36] describes sensemaking for intelligence analysis as an iterative process consisting of two loops, i.e., a foraging loop for searching and extracting information and a sensemaking loop for developing a mental model about the discovered information. Another well-known work is the *Knowledge Generation Model* by Sacha et al. [38], which outlines the analyst's role in the knowledge generation process and emphasizes the significance of seamless human-machine integration through interactive engagement with the system. Foster et al. [16] describe the information seeking as a nonlinear process that includes multiple steps such as problem recognition, source selection, query formulation, search execution, and examination of results [31]. Building on prior work, the claim-based workflow presented in this paper follows an iterative process that encompassing multiple steps, such as query formulation, search execution, and result consolidation. However, in this work, we place particular emphasis on the early stages of information discovery demonstrating how a serendipitous claim-based workflow – integrating three typical user pathways – can enhance both hypothesis generation and validation. The work by Han et al. [18], which examines how patent examiners perform a series of document-related activities, is the most closely related work to claim retrieval workflows. However, to the best of our knowledge, no explicit workflows have been proposed for analyzing large document corpora with a focus on the preliminary exploration phase of the research process.

3 Requirement Analysis

In this section, we characterize the main requirements that inform our proposed workflow. These design considerations were distilled through a series of workshops and internal discussions with experts from various domains. We describe the details of the process below.

3.1 Project Evolution

Over the course of a year, we conducted workshops with a research lab working on interdisciplinary research in engineering and medicine. Our core collaboration team was a professor, a senior scientist, and a PhD student, while additional people were invited to workshops based on expertise and availability. Monthly workshops supplemented with bi-weekly meetings over the course of nine months aimed to *discover* the right problem, *design* suitable solutions, and move towards *implementing* prototypes [39]. Through these sessions, we identified opportunities to support researchers in the early stages of their work, particularly when exploring large volumes of scientific medical literature. This iterative process yielded

a number of initial low-fidelity prototypes used to verify discussed features and requirements.

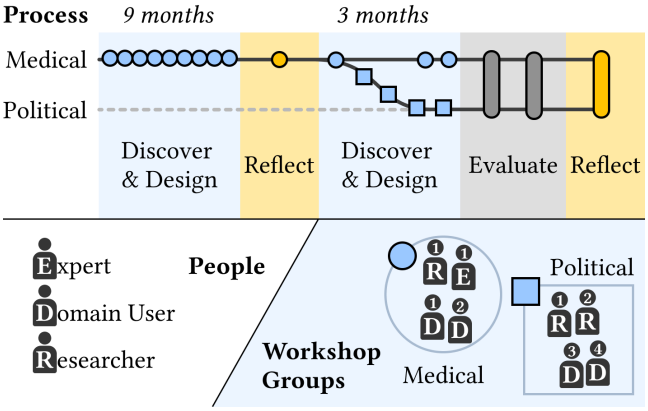


Figure 2: The overall project evolution. The initial collaboration with medical researchers was expanded to include analysts in the political and social sciences.

To investigate the generalizability of these initial findings, we expanded the scope beyond the medical research domain to include collaborators from political and social sciences. When working with text corpora, experts from these domains were deemed to explore claims from a more qualitative perspective than their quantitative medical counterparts. For example, while medical researchers are often concerned with details regarding outcomes and study design, those working on political topics (e.g., political speeches) might be interested in more subjective aspects such as the semantics of reformulations and tone. Additionally, while quantitative researchers are looking for alternatives for established systematic workflows for literature analysis, social scientists desire tools to navigate large unstructured text corpora (i.e., social media, archives, etc.) efficiently. This broader engagement reaffirmed that, although the two domains often have different analysis perspectives (i.e., quantitative vs. qualitative), similar workflow challenges exist across these domains, indicating a shared need for improved research tools and practices. However, in both cases, the target user group was domain experts as the challenges related to formulating hypotheses based on large text corpora do not naturally extend to laypersons and the general public. The complete project evolution is depicted in Figure 2.

3.2 User Needs and Challenges

We highlight our collaborators' recurring work patterns that inform the requirements for a system to improve the hypothesis exploration process. When aiming to generate novel insights through text analysis, our collaborators typically begin their investigations with an exploratory phase. This may involve skimming literature to identify the state of the art, discover gaps, or formulate new hypotheses. The initial stage of exploration is characterized by a balance between serendipitous discovery and directed inquiry, allowing experts to explore a broad range of ideas without becoming mired in any single avenue prematurely. Supporting this type of unexpected discovery is an important but comparatively neglected factor in information seeking, research, and ideation [44]. Figure 3 illustrates the stark contrast between the characteristics of early preliminary exploration and the more structured research processes that typically follow. We discussed this dichotomy with our experts. They highlighted that while the structured analysis phase involves significant effort, they are overall satisfied with the tooling support that exists and described this phase as intuitive and well-defined. For the preliminary exploration phase, however, we noticed considerable uncertainty among our collaborators — and even ourselves — regarding which tools to use and how to use them effectively. A recurring challenge was how to approach *hypothesis exploration*, encompassing both the generation and validation of new ideas.

Stage of Research Process	Preliminary Exploration	Structured Analysis
Activities	Discovery, Ideation Exploration	Consolidation Quality Assurance
Way of working	Rapid & Lean, Fail-Fast Quantity over Quality	Structured & Systematic Quality over Quantity
Artefacts	Mindmaps Unstructured Notes	Tables Documents
Number of Papers Considered	1000+	20 - 200

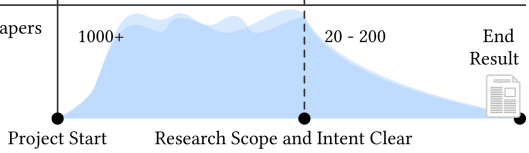


Figure 3: An overview of characteristics observed at various stages in research.

During discussions with experts in the political and social sciences, the limitations of their existing workflows were evident. Their analysis workflow often requires them to curate and compare claims from multiple sources. Often, they rely on full-text keyword searches using `CTRL+F` to locate the most relevant claims. In unstructured document collections like newspaper archives or political speeches, this approach makes searching for interesting patterns very effort-intensive. Researching discourse style is particularly demanding as it involves analyzing nuances in language choice with countless variations. For instance, there is a wide spectrum of possible formulations for 'activists', ranging from 'freedom fighters' to 'terrorists.' Neither keyword search nor close reading are appropriate for this kind of analysis. Hence, solving these challenges would require *more targeted and efficient claim retrieval* beyond a simple keyword search. Ideally, *retrieving similar claims given an example* would be automatically supported by the provided tooling. Finally, their current workflows are not scaling optimally to *large collections of text*. Experts in more quantitative fields, such as medical research, report similar difficulties in their workflows. The continuously increasing amount of information makes attempting to read academic papers in-depth overwhelming and forces users to rely on abstracts or auto-generated summaries. For instance, a common strategy to manage an immense amount of literature is first filtering the papers by skimming their abstracts for important keywords. At the same time, researchers feel a need to include as many papers as possible in their analysis to avoid overlooking important findings. While this approach might be feasible when writing large

survey papers within a long-term research project, the time and effort required are in direct conflict with the spirit of preliminary investigations. Thus, better tooling would allow the user to make an appropriate *trade-off between effort and coverage*. Additionally, studies have shown that even in peer-reviewed research, published findings are often false [22]. Consequently, reading beyond the abstract to evaluate study designs and assessing prevailing biases in the research field is crucial. This necessitates a different way to navigate papers during hypothesis validation.

Furthermore, the extensiveness of the text corpora also poses challenges during the exploration of literature for new ideas, as researchers are forced to traverse the corpora by superficial criteria such as titles, citations, and years. This method constrains experts to read only within a narrow scope of interconnected literature, creating a lack of diverse perspectives and impeding the creation of novel ideas. Therefore, both the activities of hypothesis validation, as well as exploration, would benefit from *alternative navigation strategies* that surface claims within their nuanced context more efficiently. Overall, the identified challenges highlight that users should be better supported in *working with large volumes of text* and be provided with alternative strategies to *navigate the information*.

According to the theoretical framework *HyGene* [43], individuals draw upon long-term memory, working memory, and environmental cues to form multiple candidate hypotheses during hypothesis generation. Once candidates are generated, the process shifts to hypothesis validation, where these hypotheses undergo an iterative consistency-checking process to identify and eliminate inconsistencies. New hypotheses are continuously introduced as others are discarded, emphasizing the dynamic nature of this phase. A system that supports users in hypothesis exploration should, therefore, *offer tools and abilities to aid in rapid and effective consistency-checking*. Additionally, a workflow should capture the iterative nature and flexible alternation between exploration and validation.

Across all these requirements, we facilitate deliberate user actions while also leaving room for serendipitous discoveries. To account for both of these seemingly opposing facets, we aligned with the viewpoint of Corneli et al. [11], framing serendipity as a systems' capability to increase the potential for exciting yet unexpected discoveries through intentional design choices. Specifically, alternative information traversal strategies surface information in unanticipated ways and are thus more likely to grab attention [9]. Similarly, ensuring context for claims can impact the relevance of the perceived information and thus increases the likelihood of drawing novel unforeseen conclusions.

3.3 Description of Requirements

Based on the discussions above, we formulate the following requirements for a system that aims to improve hypothesis exploration:

- R1: **Efficient Hypothesis Validation** – Support quick and iterative testing of multiple hypotheses.
  Motivation: Testing hypothesis involves assessing diverse perspectives from multiple sources. Without proper tooling support, the effort involved in manual verification within a large corpus is not practical.
- R2: **Trade-off Between Breadth and Depth** – Ability to balance between covering a large share of the available sources and providing deep, context-rich insights.
  Motivation: Breadth captures diverse perspectives, while depth ensures rigorous conclusions grounded in an accurate context.
- R3: **Claim Consistency Assessment** – Capability to identify supporting evidence or contradictions for a given claim.
  Motivation: Validating a hypothesis is best achieved by locating the most pertinent claims related to it.
- R4: **Support for Large Volumes of Text** – Ability to handle and process large amounts of text data.
  Motivation: The increasing amount of accessible information demands careful consideration of a broad context during hypothesis generation and validation.
- R5: **Alternative Traversal Strategies** – Capability to navigate immense amounts of text beyond standard information hierarchies such as titles, abstracts, and summaries.
  Motivation: Summaries provide a top-down perspective on data. However, other critical dimensions, such as secondary variables and remarks, may remain obscured within these structures. Full-text searches only capture exact matches of characters and cause tedious manual labor when applied to a large collection of documents.
- R6: **External Memory And Provenance Support** – Facilitate the organization, tracking, and synthesis of information.
  Motivation: The exploration process can easily include the analysis of hundreds of papers with relevant claims at any location. Helping the user keep track of their analysis insights, their process, and the provenance of the information is only possible through tooling support.

3.4 Discussions with Experts

The validity of the proposed requirements was verified by interviewing two independent medical researchers. The full interview protocol is available in Appendix B.1. The first expert E1 conducts reviews and meta-analyses in medical research as part of their full-time occupation. Expert E2 is a professor and researcher in the field of pre-clinical research. Both work with large volumes of research literature and actively use discovery processes within their daily work. Efficient hypothesis validation (R1) across the vast available literature (R4) was immediately acknowledged as a key challenge when shaping the scope of surveys and meta-analysis, while any support in automatic claim consistency assessment (R3) was projected to be a major contribution to the field if done sufficiently well. Both experts often find themselves skimming the full texts of papers to spot desired variables for writing meta-analyses and to identify accurate effect sizes. They clearly identify challenges in all existing literature navigation strategies (R5) supporting this process. They also note that provenance is a challenge in their daily work (R6) and that they commonly encounter information loss and additional work due to manually retracing previous findings. The trade-off between breadth and depth (R2) was intuitive to the experts, specifically when exploring less familiar literature. E2 mentions that this might be specifically relevant for students and young scientists.

4 Workflow

The following section illustrates how these requirements can be met using a claim-based workflow paired with effective retrieval strategies. Our proposed workflow is shown in Figure 4. We describe *searching* for relevant claims in the corpus (A), discuss the *contextualization* of findings (B) and demonstrate the *consolidation* of results (C). The time at which to transition between these activities depends on the users' preferences and is fully within their control.

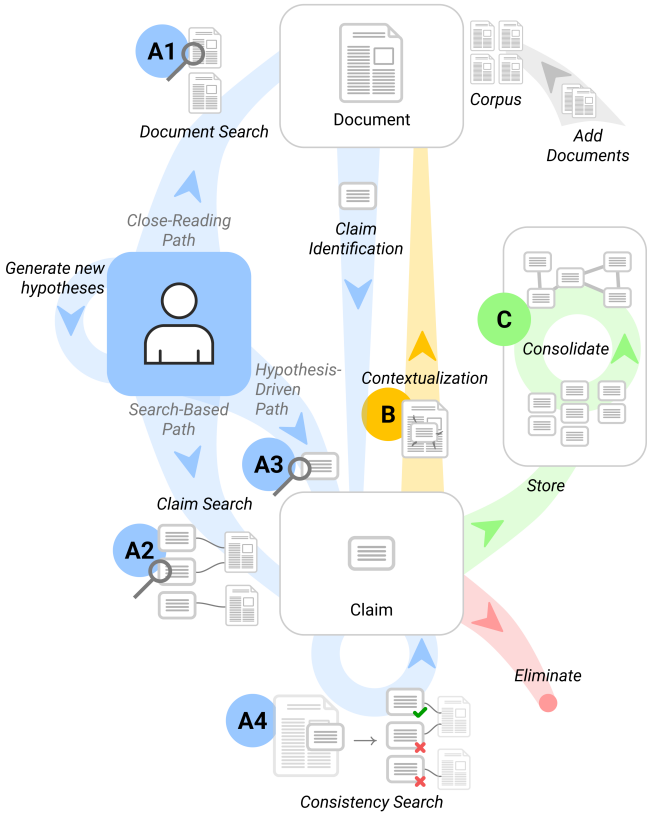


Figure 4: A claim-based workflow for hypothesis exploration in large document corpora. Users can expand the scope of their exploration by employing one of three search pathways. The workflow focuses on a claim-centered hypothesis generation and validation loop by facilitating efficient contextualization, targeted retrieval pathways, and suitable consolidation tooling.

Searching to Expand the Scope. In the workflow, users can initialize their exploration by employing one of three pathways. These represent the aggregation of common user behavior patterns encountered during the requirements formulation. Four distinct actions (A1 - A4) are envisioned to support the users following these paths. These actions can be employed in arbitrary order to expand the search scope, deepen understanding, and facilitate discovery.

Users can choose to start with close reading via document search (A1), which, outside of our conceptual framework, is often the only available option. Following the predefined document hierarchy, users identify claims through skimming and reading important sections. The keyword-based *claim search* (A2) supports users in both hypothesis generation and validation through the direct retrieval of target claims. The ability to retrieve individual statements from a large corpus enables flexible exploration of specific themes, fostering serendipitous discovery and divergent thinking. Simultaneously, users can systematically gather evidence to validate hypotheses by using targeted keywords (R2, R5). Alternatively, the *hypothesis-driven* approach (A3) allows users to bypass the cognitive load of converting their thoughts into keywords and allows a potential system to respect the full meaning conveyed by a full sentence. Therefore, users can engage more deeply with the content by directly inputting a statement or selecting an extracted claim. The retrieval algorithm then suggests related claims from the entire corpus, accelerating evidence collection for hypothesis validation (R4). Finally, *consistency checking* (A4) allows to identify entailment and contradictions between statements. Specifically, by comparing the claim of interest with topical statements from the whole document collection, the algorithm can surface the most pertinent claims to ensure coherence. This approach allows the surfacing of single, highly relevant claims from documents with otherwise overall low relevance, offering a better trade-off between breadth and depth (R2). Therefore, it automates the otherwise effort-intensive metaphorical search for the needle in the haystack (R3). Especially during preliminary investigations, where users engage in multiple cycles of ideation and validation, the proposed additions provide greater flexibility in acquiring information and offer more tailored support than existing approaches.

Close-Reading for Contextualization. Once users identify statements of interest, they organically proceed to deepen their understanding by examining the surrounding context at the document level. This *contextualization* step (B) is crucial as naively providing users with uncontextualized statements in an isolated setting can be misleading and result in premature conclusions. For example, claims may rely on previously defined terms, reference earlier ideas, and frequently lack the necessary context for complete understanding. Thus, to obtain new insights and rigorously reason about hypotheses it is vital to offer flexible and immediate access to the full text (R5) and provide tools that support capturing context and tracking provenance (R6).

Consolidation for insights. As the number of relevant ideas grows, providing external memory support for storing and managing extracted claims becomes important to assist with the increased mental load. During a serendipitous exploration of the corpus, users can store, juxtapose, and rearrange information for creative engagement. In the hypothesis validation phase, users receive assistance in structuring, relating, and reasoning about the collected evidence. Given that both processes are dynamic, with new claims continually emerging and others being discarded, it is important to focus on user control and flexible structures. This *consolidation* support (C) is seamlessly integrated with other activities to help users sustain uninterrupted engagement (R6).

Our approach is agnostic to text type, allowing any document to be dynamically added to the corpus for analysis during ongoing investigations. The workflow supports seamless transitions

between activities at any stage, accommodating the non-linear and task-dependent nature of the process.

5 Interface Design

The *NEEDLE* interface - **N**avigating **E**fficiently through **E**xtensive **D**ocument **L**ibraries for **E**vidence - consists of a central *Canvas* and a *Search Panel*, which form the default workspace (Figure 5). A *Corpus View* and *Close-Reading View* open up on demand to help contextualize information.

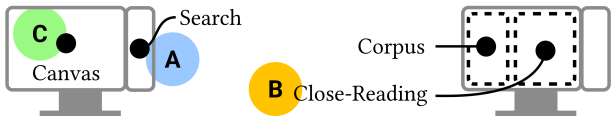


Figure 5: A schematic depiction of the components in our *NEEDLE* interface.

The interface was refined iteratively based on feedback from our collaborating medical experts. Although we describe the different views in detail separately below, they should be thought of as an integrated whole. The guiding design principles were to provide users with full control and enable seamless transitions between all panes for a non-disruptive experience.

5.1 Search

The *Search* interface follows the standard practices of information retrieval systems, where users provide keywords and receive a curated list of claims for exploration (Figure 1.1). The retrieval process utilizes named entity recognition (NER) models[3] to match entities from user inputs with those indexed in the corpus, employing appropriate ontologies[4] to retrieve semantically meaningful claims that go beyond mere lexical similarity.

In the *Search Panel*, claims can be displayed individually, ranked by their overall relevance, or grouped by document, providing an overview of the most important sources. Users can open up the full text in the *Close-Reading View* (Figure 1.2), by clicking on the claim, to examine it within its original context for deeper understanding. Alternatively, claims can be directly stored on the *Canvas* (Figure 1.3) for later analysis and consolidation.

During the search process, users can toggle the *Corpus View* (Figure 1.4) as needed, offering a comprehensive overview of all documents in the system. The *Corpus View* offers basic filtering options to restrain the search to a subset of all available documents and automatically marks inspected papers as visited, helping the user to keep track of the search process. Additionally, the encoding of the retrieval results on top of the document layout offers further insights to contextualize the search, by highlighting the distribution of relevant claims in time (R4, R6).

The overall user interface layout is meant to support two main configurations. Users aiming for broad literature coverage can first utilize the default interface configuration of *Canvas* and *Search Panel*, and begin collecting claims to gain an overview first. In contrast, users engaged in thorough analysis can emphasize contextual understanding by examining the full text and leveraging the *Corpus View*. Users can easily alternate between these approaches (R2).

The interface enables users to search for relevant content by entering a sentence, thus operationalizing the *hypothesis-driven path* from the workflow. This functionality supports both scenarios: selecting a previously extracted claim to gather additional evidence, or initiating a new search using their own formulated hypothesis.

For users interested in consistency checking (R3), the system employs natural language inference (NLI) algorithms to generate a ranked list of results, ordered by the extent of supporting and conflicting claims found in the corpus. We use a RoBERTa large model[5] for NLI and evaluate it on the most promising candidates retrieved by the NER-based retrieval. For further details regarding the implementation, refer to Appendix A. This approach seamlessly integrates automation to enhance user efficiency by automating the labor-intensive tasks of scanning and ranking relevant content. Meanwhile, it preserves user control at key decision points by allowing users to steer the exploration process and contextualize their results. The usage of these features is demonstrated in Figure 6.



Figure 6: Users can effectively pair techniques A3 and A4 within the interface to retrieve highly relevant suggestions.

Efficient hypothesis validation (R1) is facilitated by the interplay of multiple requirements and their tight integration within the workspace. Semi-automatic claim consistency checking (R3) and alternative traversal strategies (R5) enable the effective identification of necessary evidence. The system supports large volumes of text (R4) while allowing users to balance breadth and depth (R2) according to their needs. This flexibility allows users to calibrate

[3]Medical instance:https://lhncbc.nlm.nih.gov/ii/tools/MetaMap/run-locally/MetaMapLite.html, political instance:https://spacy.io/models/en#en_core_web_sm
[4]Medical instance:https://www.nlm.nih.gov/research/umls/index.html, political instance:https://wordnet.princeton.edu/

[5]https://huggingface.co/FacebookAI/roberta-large-mnli#model-details

the rigor and contextual detail of their hypothesis validation, considering factors like domain-specific risks and the importance of accuracy at their current stage of analysis.

5.2 Close-Reading

The *Close-Reading View* is shown in Figure 1.2. We intentionally keep this view simple and display the plain full text without additional structure and support to incentivize users against a document-centric workflow. Users can mark relevant statements in the text and instantiate them into claims. They can also add relationships between marked text snippets, supporting their reasoning process and allowing them to externalize their understanding of important sections. This facilitates efficient contextualization while automatically recording provenance.

5.3 Consolidation

With the ability to capture a large volume of contextualized claims from different sources, it is crucial to support the user in managing the accumulated information effectively. The *Canvas*, see Figure 1.3, offers a whiteboard-style interface where each collected claim is instantiated into its own card and can be placed freely on an infinite canvas. These claims can be directly added from the *Search Panel* or during close reading. Customizable tags can be added and used to automatically re-organize the spatial layout, aiding users in reasoning about the data from multiple perspectives. The relations introduced in the *Close Reading View* are equally supported within the *Canvas*. This allows the user to start from unstructured collections of information and continuously reorganize the content for sensemaking [13]. This support ranges all the way to creating complex diagrams, an established technique for reasoning about complex arguments [46]. The provenance of each claim is automatically tracked, and users can always refer back to the source document through a single click, allowing for a seamless transition between the *Canvas* and the *Close-Reading View*. We choose an infinite canvas as visuospatial thinking helps users to reason effectively about the collected content and is suitable for creative processes [47]. Additionally, workflows involving creativity benefit from the ability to create and remove artifacts iteratively, continuously restructure information, and evaluate intermediate results [33, 48].

In summary, our interface offers a comprehensive instantiation of the proposed workflow while maintaining conceptual simplicity. With four consistent views in fixed positions, the design ensures users can easily navigate and orient themselves within the tool.

6 Use Case

In this section, we present a use case to demonstrate how the proposed workflow can be applied using the example *NEEDLE* interface. Although the use case is narrated under a hypothetical scenario, its design is inspired by some of the users' actions and feedback gathered during the user study (see section 8).

6.1 Case Description

Hypothetical Scenario: ***Russia's Framing of the Ukrainian War.*** After reviewing the results of a study about how Central Asian countries utilized sovereignty frames in their United Nations General Assembly speeches following the Russo-Georgian War, Crimea Annexation, and the Russo-Ukrainian War [1], a political expert seeks to extend their understanding of the topic and gain additional qualitative insights.

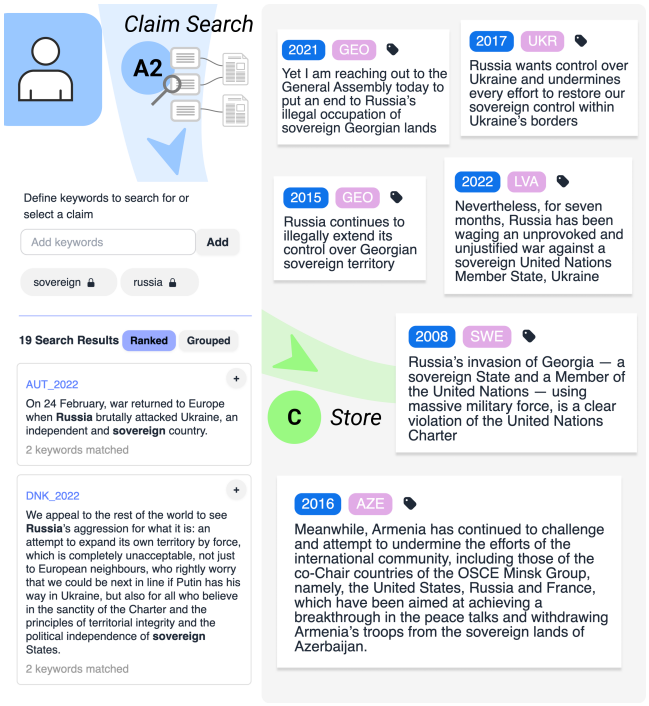


Figure 7: The expert explores claims with the terms *'sovereignty'* and *'Russia'* utilizing the *search-based path.*

Search-Based Path. Motivated by the study results [1], the political expert aims to explore which countries mention the term *'sovereignty'* in relation to *'Russia,'* as this can uncover how nations perceive and frame Russia's actions in the context of international law and geopolitical conflicts. Consequently, the expert follows the *search-based path* and executes the *claim search* with the two keywords, as shown in Figure 7. The search leads to 19 matching claims. After skimming through the results, the expert stores the most interesting claims on the canvas. In turn, the expert notices that Azerbaijan discusses Russia as one of the countries that aims to achieve a breakthrough in the peace talks between Armenia and Azerbaijan. Russia mentioned in the context of peace negotiations motivates the expert to propose a hypothesis suggesting that Russia advocates for peace.

Hypothesis-Based Path. To validate this hypothesis, the user transitions from the *search-based path* to *hypothesis-based path* and inputs the following statement in the *Search Panel*: *'Russia wants peace.'* The expert uses the *consistency search* feature to access supportive and disputative claims. The results show that many European/Western countries dispute this statement. However, there are also supportive claims that are of particular interest to the expert. While the majority of these claims come from Russia itself, speeches by other countries, such as Bolivia and Palestine (Figure 8), also offer statements that back the hypothesis of Russia contributing to peace in various geopolitical situations.

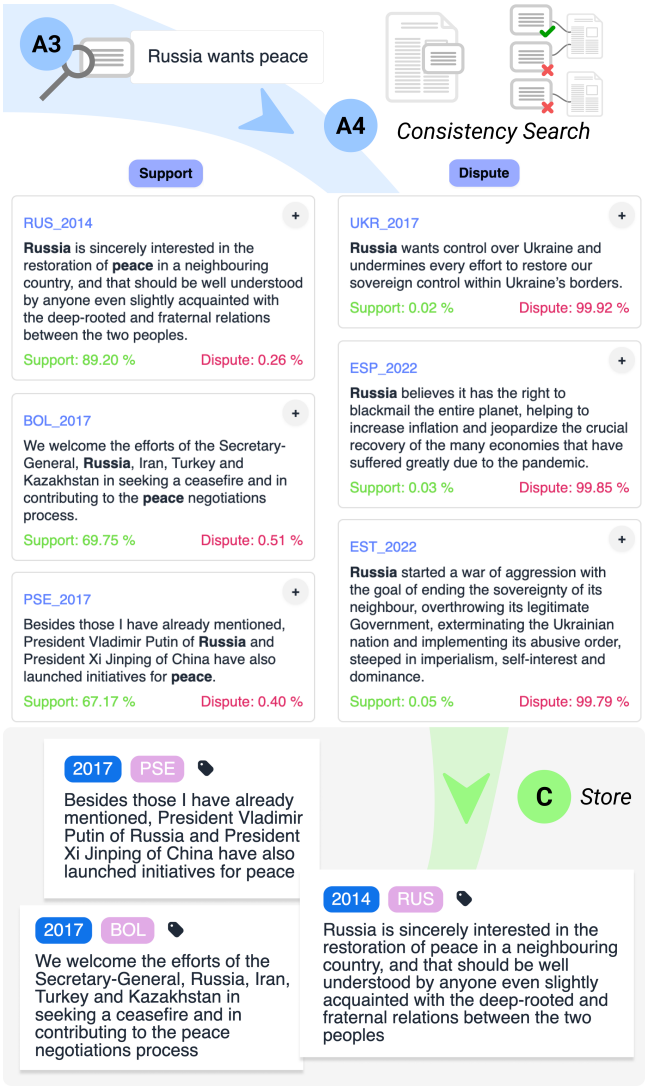


Figure 8: The *hypothesis-driven path* is used to assess claims supporting and disputing the hypothesis specified by the expert.

Contextualization. The expert finds Russia's claim of being interested in the restoration of peace in a neighboring country worth further inspection. The expert uses the *Close-Reading View* to understand in what context this statement was made. Apparently, this statement refers to the Maidan that took place in Ukraine in 2014. As shown in Figure 9, Russia blames the Ukrainian army for the humanitarian disaster in south-eastern Ukraine. The expert assigns this statement as context to the particular claim and it gets added to the canvas.

Serendipity. The expert seeks to identify additional justifications for Russia's invasion of Ukraine. To gather this evidence, the expert executes a chain of *claim searches* by directly selecting an already extracted sentence and accessing similar claims suggested by the retrieval algorithm, as described in section 4. The system is able

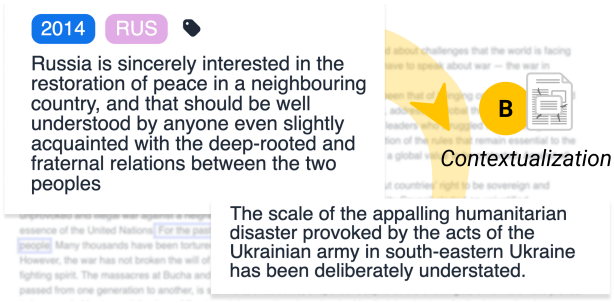


Figure 9: *Close-Reading View* is used for contextualization.

to retrieve results with highly varied language, a capability that enables unexpected findings, hence supporting the user in serendipitous discoveries. This also demonstrates how intentional action and serendipity are not mutually exclusive. As showcased in Figure 10, combining *claim search* with *consistency search*, the expert observes further accusations Russia makes towards Ukraine and *'Kiyiv's neo-Nazis,'* NATO, and Western countries in general.

Consolidation. Finally, the expert organizes the collected claims on the canvas, arranging them by the years the speeches took place and marking the relationships between the claims using edges. An excerpt of the consolidation step is shown in Figure 11.

6.2 Analysis of Claim-based Workflow

Based on the use case described above, we can demonstrate that the proposed claim-based workflow was utilized in the following way. During the exploration, the expert employed all the paths apart from the *document-driven path* since they had no prior knowledge regarding the document corpus. The *search-based path* was used to expand the search space with the aim of promptly identifying relevant claims that were then later contextualized. The *hypothesis-driven path* was used for more tailored identification of relevant claims and hypothesis validation. The *consistency search* function was used to narrow down the expanded search space to only supporting or disputing claims. Lastly, the *Canvas* enabled the *consolidation* of the found claims helping to visually keep track of how the hypothesis is evolving in the process. The presented use case illustrates a small example of the knowledge discovery process. In a real-world scenario, the various paths would be executed multiple times.

7 Expert Interviews

As part of the evaluation, we conducted interviews with the same medical experts we initially interviewed to validate our requirements in section 3. Our goal was to gather feedback about our workflow and the *NEEDLE* interface focusing on their strengths and weaknesses and their relevance and potential within the field of medical research. We walked our experts through the tool while demonstrating our methodology within the proposed workflow and provided them the opportunity to offer feedback on both the workflow and the interface. Both experts recognized the potential of the workflow when starting a new research project. Especially for less experienced researchers, the option to explore the literature using a hypothesis-driven approach and being able to see in which

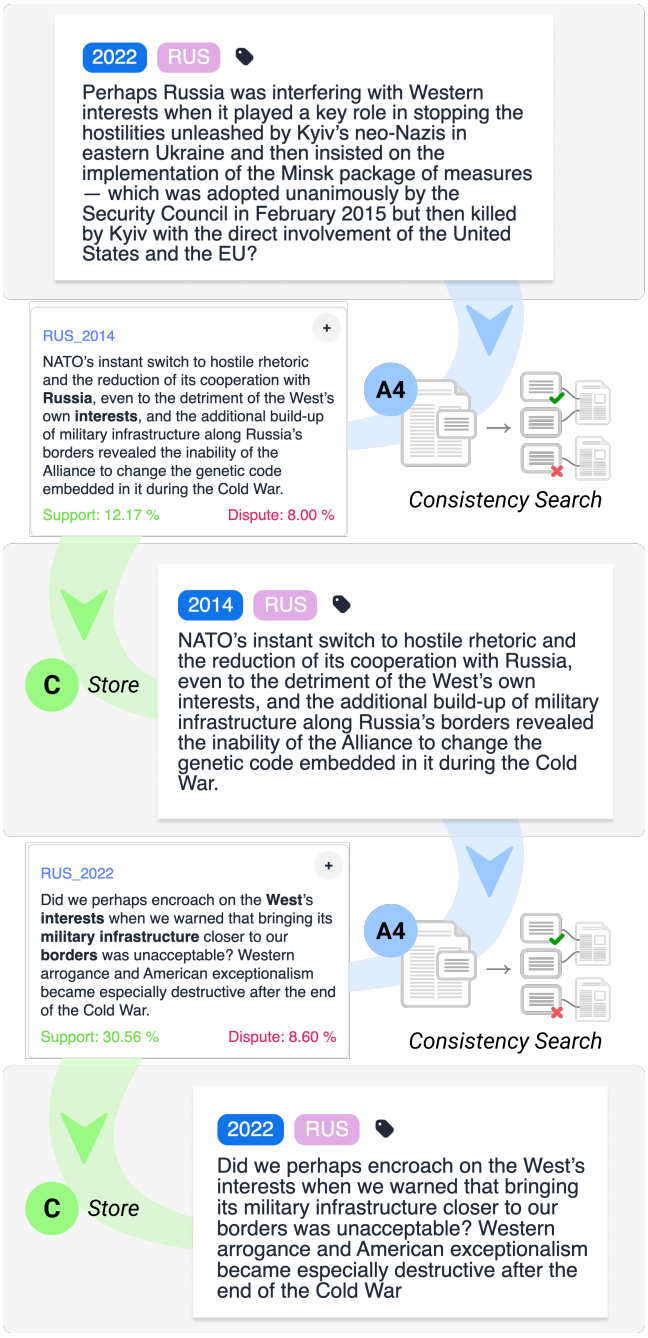


Figure 10: The retrieval algorithm supports accessing relevant claims, hence supporting serendipity.

communities certain claims are surfacing can be helpful to guide further exploration. E2 noted that a common issue in medical research is the prevalence of strongly entrenched hypotheses within certain communities, leading to biased experimental setups and the omission of key evidence from other research areas. They noted that the tool has the potential to improve this process and support researchers in exploring scientific literature along unfamiliar paths.



Figure 11: The expert organizes the retrieved claims on the canvas according to their relevant properties (e.g., year).

E1 highlighted that augmenting our workflow with automatically extracted statistics from documents, while still maintaining a high level of user control, provenance, and the unique similarity search, could also be a valuable contribution to the medical field. Further, E1 mentioned that our nuanced consistency search feature goes beyond the aggregated summaries provided by tools like Consensus, as the missing support for manual verification and contextualization is making AI-generated summaries unsuitable for medical research. Overall, E1 summarized that due to associated risks in his domain, having full control over the interpretation of results and being in control of steering the analysis process is a crucial requirement. E2 identified the feature for support and contradiction search as highly applicable and proposed to test the tool within a classroom setting. In their forthcoming lecture they plan to address the issues of contradicting research in medicine with young scientists, however, finding suitable examples is currently difficult and typically done through extensive manual labor.

Overall, the experts described the workflow as an intuitive formulation of implicit behavioral patterns they observed in the field. It establishes a structure for preliminary explorations in research and they see potential in various implemented features to support medical researchers in their work.

8 User Study

This section presents a user study to evaluate the efficiency and usability of the proposed interactive interface. We capture the participants' experiences using the interface during hypothesis generation and validation within the two example contexts of medical research and political discourse. To ensure systematic analysis and comparison, a baseline is established from users' existing workflows and preferred choice of tooling.

8.1 Study Design

Participants. In this study we recruited a total of ten researchers (aged 27-35) all of whom confirmed to have experience in performing literature reviews and working with large corpora of text. The participants featured backgrounds related to either medical research (e.g., clinical research, neuroscience, biotechnology) or political discourse (e.g., economic development, public policy, international relations). Five researchers were sourced from each of these two target domains. In the following, researchers from the medical domain are labeled as M1-M5, while those from the political domain are labeled as P1-P5. During the initial selection, all participants acknowledged that they had previously utilized various workflows and tools to carry out exploratory research.

Process. The study began with obtaining formal consent from the participants, followed by an initial discussion and introductory video outlining the study's goals and agenda. After this, the participants were introduced to the *NEEDLE* interface using a demo video and then allocated time to familiarize themselves with the interface and ask any clarifying questions regarding the functionality. Before the study, the experts were also asked to prepare the tooling (e.g., note-taking, literature management) they would normally rely on for these types of hypothesis-generation tasks. In the main phase of the study, depending on the domain to which they were allocated, the participants were required to solve two separate but comparable 15-minute tasks with our *NEEDLE* interface and their own baseline tooling. This chosen setup enabled both a slow, more thorough study participant to go through most subtasks, while making sure a fast, more superficial user would have enough remaining tasks to not stay idle during the task solving stage. The order in which the tools were used was alternated between participants to account for the inherent learning that occurs when working with a corpus of information. For the medical domain, the tasks involved finding correlations between different diets and diseases from a sampled collection of 3000 open-access medical articles while the political use case focused on analyzing different global conflicts using a corpus of UN speeches from the years 2008 to 2022[6]. Before starting the tasks, we also gauged the participants' prior knowledge by letting them self-assess their experience regarding the task topic. Throughout the study, the users were actively encouraged to think aloud about their approach while performing the tasks. Once the two tasks were completed, the participants filled out a questionnaire to provide feedback on the usability and effectiveness of both sets of tooling. After this, we also conducted semi-structured interviews with the participants to understand their overall experience using the interface, including the reasons behind their actions, the usability limitations of the interface, and their suggestions for improvements. To guide the discussion, we asked for feedback regarding the main components of the interface, namely, the *Search View*, *Close-Reading View*, and *Canvas*. Additional details can be found in the complete user-study protocol included in Appendix B.2.

[6]https://github.com/sjankin/lancet/blob/master/report2023/UNGA_indicator/data/ungd_files.RData

8.2 Study Results

Based on our observations and the collected quantitative and qualitative feedback, we outline the key insights gained in the study. Particular emphasis is on comparing the users' workflows for generating and validating hypotheses with our *NEEDLE* interface versus their existing baseline tools. The tool's efficiency and usability, general qualitative feedback, and the participants' suggested improvements are also discussed. The baseline was determined on an individual basis as each user was allowed to use any variety of tool combinations. In our study, the most popular note-taking applications were standard document editors (e.g., Google Docs, Microsoft Word) while sources were tracked with a more varied set of frameworks (e.g., Mendeley, Zotero, EndNote).

8.2.1 *Workflow Analysis.*

Carefully observing how users interacted with both our *NEEDLE* interface and their tools for the different tasks offered valuable insights into their workflows.

NEEDLE . Most of the participants (9 out of 10) from both domain groups began to expand their initial understanding along the *search-based path* by entering relevant keywords. The only exception (P3) engaged with a *close-reading path* to confirm an initial hypothesis from an exact political speech they knew to be useful. It was observed that many users (7/10) tended to first iteratively expand the scope by searching for a lot of different claims and storing them on the *Canvas* (A2, C). Only after retrieving numerous claims did participants proceed to consolidate the ideas by establishing connections between separate statements (C) and investigate the context in the full documents (B). The rest of the experts were more dynamic in their use of the tool and frequently transitioned between searching for new claims (A1, A2) and consolidating their findings (B, C). Additionally, some users (M3, P1, P2, P4) remarked that due to the time limitations, they are actively forced to prioritize either covering the entire task or doing only parts of it thoroughly. Medical experts M3 and M5 also lamented the lack of time, emphasizing the importance of identifying contradictory evidence in their workflows, but noting that they were unable to do so due to time constraints. Keeping track of how participants maintained and assembled their claims on the *Canvas* provides insight into the range of methods individuals use to contextualize and consolidate information. Some (P1, P2, P3) relied on basic meta information such as year and source document to group claims on the board while others (M1, M3, M5) used the connection functionality to build complex mind-maps to reflect their reasoning and mental process.

The rest simply sorted the claims by the approximate similarity between statements. One user (M3), however, eventually regretted this unstructured approach and complained about needing to spend time re-tracing their thought process. A general trend was also observed depending on the backgrounds of the participants. Those with prior knowledge regarding the tasks were more likely to engage with hypothesis validation while those less familiar with the topic focused on generating new ones. Depending on the task domains, we noticed some differences as well. Participants working in the medical domain were much more prone to rely solely on the keyword search. Conversely, those engaged with the political speeches also made effective use of search filters and corpus views to isolate specific information. Some political experts (P4, P5) even mentioned that due to the vagueness of the task description, it is slightly difficult to choose which keywords to begin the search with.

Baseline. When working with their own tooling, participants predominantly committed to a close-reading approach with a focus on skimming across titles and abstracts. The default workflow observed involved copy-pasting interesting ideas into a text editor and noting down the source documents using URLs or saving them into a literature management tool. Some users (M2, M5, P4) consistently complained about the need to transition between various tooling and the difficulty in retracing their thinking from previous steps. However, it appeared that familiarity with this way of working meant that most users intuitively knew how to perform these steps even if they were not necessarily very efficient. The participants in the studies that had first solved a task using our *NEEDLE* interface vocalized how they are now more conscious of mundane actions such as copying sources or writing up metadata regarding the found claims. Nevertheless, not everyone (M3, M5, P3) felt the baseline approach was inherently worse for these 15-minute tasks. One expert (M5) who had prior knowledge regarding the topic said *"This task is too simple and thus can be solved good enough by just reading abstracts but in a contained setting, if the question was more specific, then manually working through documents is very ineffective."* Another (P3) echoed this sentiment by saying that *"I can see why it (the baseline approach) would be annoying if I did not know anything about the topic but for me it makes sense here."*

8.2.2 Qualitative Feedback.

The general qualitative feedback provided by the users throughout the study is compiled below.

Structuring Thoughts on the Canvas. There was consistent appreciation from the users regarding the *Canvas*. Several experts (M2, P2, P5) liked the flexibility of moving around claims and having a consistent visual representation of how their argument is being incrementally constructed. P4 said *"The canvas is very intuitive and easy to work with"* while P5 added, *"It really helps me make sense of my ideas."* User P2 aptly summarized that *"the best part is how well it scales to building full and complex arguments."* Others saw possible utility in other contexts as M5 said *"I would even use it to just organize all my papers"* then adding that *"Having the claims makes it easy to remember why I save it in the first place."*

Smart Search. Participants (M2, M4, P2) appreciated how fast they could look through many claims. M3 specifically highlighted the function to group claims by documents and how *"It is good to click through all the sentences in the text without scrolling the entire doc."* M2 and P5 were positively remarking how the system also retrieves matches of synonyms, as they were expecting a simple full-text search. M1 liked what they referred to as *"shotgun search,"* being able to provide all keywords at once and retrieving many potentially relevant claims to quickly get an overview.

Integrated Layout. Another design choice that generated a lot of discussion was the integrated layout. Most users (8/10) made some mention of how it is very convenient to have all the functionality and tooling compactly in one place. The usefulness of this aspect was emphasized by the participants with medical backgrounds as their standard workflows usually involved using many different tools in parallel. M3 and P2 were also impressed by the ease at which it is possible to transition between the different views, with quick access to the full texts being a particularly useful feature. Participant M3 added that *"It is very good that I can easily see the full text again because I had forgotten why I initially saved this claim."*

8.2.3 Efficiency and Usability.

We assessed the subjective experience of our participants by asking them to rate their agreement with each of the statements shown in Figure 12 on a 5-point Likert scale. Overall, the users found the ability to identify relevant passages across papers (4.10±0.74 vs. 1.70±0.82) and tracing back information (4.20±0.92 vs. 2.00±1.05) as the main strengths of our tool compared to the baseline. More participants felt able to explore a wide range of claims with our *NEEDLE* interface (5/10), than with their own workflow (1/10). Users generally found it easy to generate exploration ideas and determine subsequent steps within the constraints of our task setup in both tools. The questionnaire further indicates that users subjectively had to invest more effort and felt more time pressure using their own baseline system compared to the *NEEDLE* interface. Their ratings on their performance in the task, and the effort required to reach it did not differ much between the baseline and our tool.

8.2.4 Desired Improvements.

We collected the participants' feedback on the desired improvements as follows.

Advanced Search. Several participants (M2, M3, M5) from the medical domain, in particular, missed an advanced search option. They felt that using even more contextual information, such as authors and paper types, makes it easier to utilize existing knowledge and intuition regarding the domain. They remarked that they are used to fine-grained control over their searches, selecting the section of the paper that is being matched and using boolean logic to formulate complex queries. While over-reliance on familiar patterns for hypothesis exploration may hinder the serendipitous discovery of novel insight, the addition of such search conditions appears to present a valid compromise.

Improved Consolidation. In the interviews, the users raised concerns regarding scaling and opportunities for managing the ever-expanding *Canvas*. Currently, the maintenance of the individual claims is manual to ensure complete user control. The participants were worried that as the number of claims on the board grows, they would like to see all claims listed in a table view, which is where

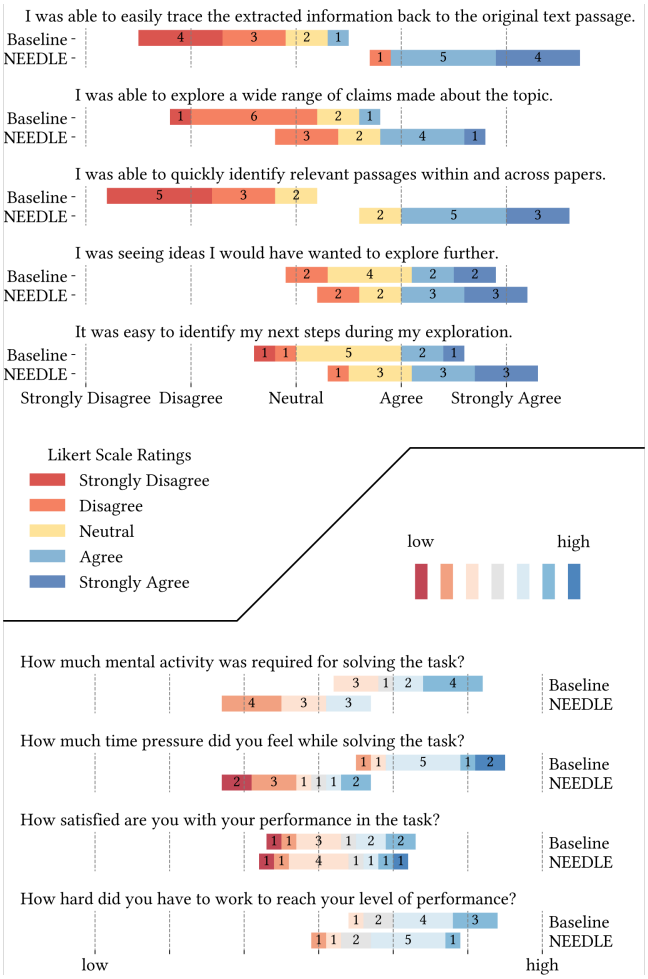


Figure 12: Subjective ratings from our ten participants regarding their experience in five key aspects of the task (top) and four questions to measure the task load (bottom), comparing the *NEEDLE* interface to their chosen baseline.

they would naturally move to as the analysis progresses. Another user asked whether they could open a separate *Canvas* to divide the information into different spaces. M5 and P3 particularly liked the idea of adding Post-it notes with their own ideas to complement the extracted claims. Others suggested opportunities for improvement were providing automatic clustering methods using the semantics of the claims beyond the simple tag-based sorting our tool provides.

Contextualization Support. The initial version of the interface consciously did not import the paper structure into the *Close-Reading View* to motivate users to navigate beyond the existing information hierarchies and ingrained workflows. However, many users, especially for the medical use case, lamented that this structural information is essential to quickly identify sections where close reading might be useful and those where it is certainly not needed. Thus, re-introducing general paper structure to the raw texts in the interface might help merge the strengths of existing workflows and our proposed approach. Similarly, aligning with these existing structures makes the tooling more familiar and intuitive.

9 Discussion

In the following section, we consolidate the discussion of results from the use case, expert interviews, and user studies, organized by insights related to the workflow and the *NEEDLE* interface.

9.1 Workflow

The workflow formalization was considered intuitive by the experts interviewed in section 7, who acknowledged its potential value across various areas of medical research. They highlighted its capacity to assist in bridging disciplinary gaps, particularly in the discovery phase of research projects. Additionally, when writing medical surveys and meta-reviews, experts identified promising synergies between this workflow and systems that automatically extract relevant evidence and statistics. One expert also noted its value for literature analysis in classroom teaching settings.

Our use case further demonstrates the workflow's flexibility in systematically providing an overview of existing claims, utilizing hypothesis-driven searches to retrieve targeted results, and serendipitously chaining multiple searches to uncover novel insights. This variability in inquiry modes facilitates exploring literature along new pathways, as illustrated in Figure 13.



Figure 13: The different traversal strategies enabled by document search A1 and claim search A2 & A3.

Overall, the workflow enhances research by providing a flexible approach for exploring literature, generating hypotheses, and uncovering interdisciplinary connections. By integrating dynamic consolidation of insights and advanced NLP-based consistency checking, the approach supports researchers in navigating complex information landscapes and efficiently gaining valuable insights.

9.2 Interface

Our system can effectively support complex workflows for hypothesis generation and validation. Using NLI to discover supportive claims significantly enhances hypothesis validation, as demonstrated in the use case. The ability to balance breadth and depth through a combination of consistency search and hypothesis-driven exploration encourages curiosity and accelerates the discovery of relevant themes in large document corpora. While the NLI model performs well for claim consistency assessment, there is potential for improvement in synonym extraction by employing more specialized embedding models. The system's capacity to handle large volumes of text is generally robust, though user feedback indicates a need for better methods to structure an abundance of claims on the *Canvas*. Alternative traversal strategies, such as consistency search, prove effective in uncovering hidden correlations and insights across documents. Additionally, while the current support for

external memory and provenance is sufficient, there are opportunities to enhance the tracking of executed steps for more advanced use cases. Keeping track of the order in which users access documents and store claims maintains an overview of the process and allows users to retrace their steps. Overall, our approach effectively addresses a real-world need for a flexible and efficient workflow for hypothesis generation and validation, providing valuable tools for users engaged in the preliminary exploration of literature.

9.3 Limitations and Future Work

Our initial results demonstrate the usefulness of our proposed claim-based workflow for hypothesis generation and validation. We do, however, acknowledge that there are limitations that should be addressed in future work.

While the presented user study provided valuable insight into how users adopted our workflow in task settings using the *NEEDLE* interface, it also confirmed the need for additional evaluation. It was evident that hypothesis generation is a complex process that is difficult to capture in isolated settings due to the variety of user behavior and various domain-specific requirements and nuances. Therefore, future work will focus on observing user workflows over extended periods of interaction under constraints that better reflect practical applications. The current work focuses on the medical research and political science domains, however, we believe that the workflow will perform well in related fields. Further studies are needed to confirm generalizability.

Another avenue for future work involves the inherent scalability limitation of our machine-in-the-loop workflow. Users are bound to explore ideas one at a time, and the consistency checking support is only run when explicitly instructed to, limiting it to a narrow scope of the analysis. While the current tool is intentionally designed to prioritize user control, incorporating guidance, and allowing an AI-agent to steer the analysis process presents promising opportunities to optimize the workflow. Involving human-AI collaboration might be used to automatically surface thematically relevant contradictions found in related papers and guide the user to more interesting passages within the corpus based on their interactions [20]. Additionally, there is ample opportunity to provide further contextualization support in future solutions. For example, one approach utilizes LLMs to edit retrieved claims such that they are interpretable without any additional. context [34].

Currently, the proposed solution is solely focused on augmenting users' workflows and generating personal insights. However, there is often a necessity to externalize this learned information and convey the generated hypothesis to other involved parties. Thus, extending the interface to support users in sharing and distributing their findings in a structured way would increase the usefulness of our tooling in the overall research process. Facilitating efficient knowledge externalization and information sharing could also provide the foundation for expanding the proposed workflow to accommodate multi-user collaboration.

10 Conclusion

This paper introduced a novel workflow to facilitate efficient hypothesis generation and validation during the preliminary exploration phase of vast text corpora. The workflow unifies three typical user pathways and adopts a serendipitous claim-based approach by operationalizing sentence-based retrieval combined with effective contextualization and provenance tracking. We exemplify the proposed workflow by instantiating an interactive interface that integrates claim retrieval and consistency checking using NLP techniques. The use case, expert interviews, and user studies demonstrate how the proposed workflow enables users to traverse literature corpora in efficient ways, enabling them to generate new insights about the explored data.

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A Implementation Details

In this section, we provide technical details on the NLP components within the workflow described in section 4. Notably, the workflow is designed to be flexible, allowing many NLP components to be replaced with more suitable alternatives depending on the specific instantiation and domain.

Literature Corpus (Medical Domain). We first built a corpus of medical publications. To this end, we extracted a corpus from PubMed, a literature service provided by the National Center for Biotechnology Information in the United States, based on keyword searches for the medical topic of interest: mental health and nutrition. PubMed offers an API[7] that allows the download of full-text articles along with some metadata in the BioC format [10]. This resulted in an initial corpus of around 2,700 relevant papers. As preprocessing, citations and figures were removed from the full text, while information like authors, citations, year, and additional metadata were stored within the data structures. To enhance the dataset with citation counts for each article, we queried the Crossref API[8]. To obtain a standardized version of the venue information, another NCBI API was utilized[9]. After filtering out articles with missing metadata or other data quality issues, 2,581 articles remained. We ran a NER model linking the biomedical entities in the text to UMLS identifiers. Each document, sentence, and detected entity was finally saved in a relational database for later use.

Literature Corpus (Political Domain). For the UN debates corpus, we used a curated dataset collecting all speeches from 2008 to 2022 (2904 documents in total). These were plain text files, and the processing pipeline employed a simpler approach: for NER, we used the `en_core_web_sm` model from spaCy[10], and, in case the entity was not known, stemming on nouns as a replacement for the NER model. This way we could retrieve well-known entities from the English language like *United Nations* as well as find sentences based on simple nouns like *conflict*. Finally, each document, sentence, and detected entity was saved in a relational database for later use.

Retrieval. The way we previously acquired and stored data goes hand-in-hand with the retrieval pipeline we use for the claim search. Beyond keyword-only lookup, the retrieval was further enhanced to lookup relevant identifiers, related terms, and synonyms, e.g., using the UMLS API[11] in the medical domain. This results in higher recall in the retrieval as we used any of these identifiers to match entities in our corpus. We then matched these identifiers with the entities saved in our database. For each match, we retrieve the sentence in which the entity was found and return it to the querying user. If the search is done using multiple keywords, we rank sentences by the number of matching entities. The overall processing and retrieval pipeline is shown in Fig. 14.

[7]https://www.ncbi.nlm.nih.gov/research/bionlp/RESTful/pmcoa.cgi/BioC_json/
[8]https://api.crossref.org/works/
[9]https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esummary.fcgi
[10]https://spacy.io/models/en
[11]https://documentation.uts.nlm.nih.gov/rest/home.html

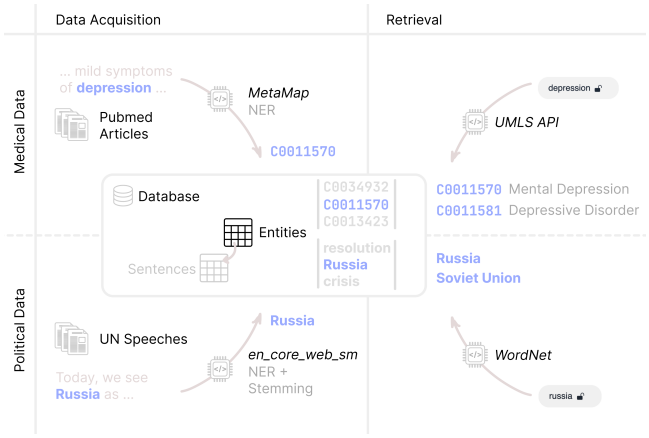


Figure 14: A schema of the processing and retrieval approaches used for both, the medical and political use case. While the solution for the medical dataset is based on unique UMLS identifiers, the political dataset relies on a more generic approach based on stemming. The keyword search is retrieving sentences by matching against the entities detected in each sentence.

Consistency Checking. For the consistency checking module, the system automatically uses the entities present in the sentence selected by the user (which are known from the indexing stage) to start a new search. This will retrieve a list of related claims as described above. Between each base claim and its related ones, we evaluate the degree of agreement or contradiction. We model this as a natural language inference task (NLI) task, where textual entailment and contradiction may be a reasonable proxy of the agreement and disagreement of claims. To this end, we used pre-trained NLI models `roberta-mnli`[12], which may be interchanged with other NLI models or LLM prompting with potentially marginally different results. This scoring then allows users to filter or group claims based on this semantic relationship.

B Evaluation Details

B.1 Protocol 1: Expert Interviews

The expert interviews were carried out in two separate stages with the same set of participants. In the first stage, we discussed the requirements for efficient hypothesis exploration, and in the second we had another discussion to evaluate our proposed workflow by presenting the NEEDLE interface. Both sets of interviews followed a semi-structured format, with each session lasting approximately one hour. The study utilized a *Miro* collaborative sketch board to facilitate the sharing of figures and note-taking. After filling out the consent form, the experts were guided through the following interview parts:

Part I: Approach, Expectations, and Prior Knowledge. (60 minutes). In the initial stage, we introduce the problem setting focusing on

[12]https://huggingface.co/FacebookAI/roberta-large-mnli

understanding experts' current workflows, as well as their expectations. We then give a brief overview of our proposed requirements and discuss how to best provide support during *hypothesis exploration*.

(1) Introducing the *hypothesis exploration* setting.
   - First ask them how they navigate the initial stages of a research project.
   - Show them Figure 3, and ask them how well it aligns with their understanding.
   - Provide them a running example of creating a survey on the effects of ketogenic diets on health to make sure the setting is clear.
(2) Understanding the experts' workflow during *hypothesis exploration*.
   - Understand how they navigate through publications when looking for specific claims.
   - Understand how experts organize and keep track of interesting papers.
   - Understand how they extract information and take notes.
   - Understand how experts trace back information to the passages in the original notes.
(3) Present our requirements and collect feedback.
   - Discuss our proposed requirements.
   - Compare how the experts' described workflows align with our suggestions.

Part II: System Demonstration and Evaluation. (60 minutes). In the second stage, we introduce our novel workflow through a system demonstration. The objective is to gather feedback on the proposed solution and discuss its strengths and weaknesses while assessing its relevance to the domain-specific challenges faced by the experts.

(1) Demonstrate the NEEDLE interface
   - Keyword search: searching for claims linking a ketogenic diet with depression and sleep effects.
   - For a claim of interest, search for support or contradiction.
   - Close-reading to contextualize the claims.
   - Show the integration of other features, such as provenance tracking and corpus search.
(2) Get Expert Feedback
   - Get open-ended feedback about the experts' impressions.
   - Reflect on experts' expectations and the self-described shortcomings in their workflow described in Part I.

B.2 Protocol 2: User Study

The user study was designed as a within-subject study, where the baseline system was compared to the proposed workflow with two different tasks within the users' domain, i.e., medical research, or political discourse. We counterbalance starting order and task effects (i.e., we split each domain group into four groups, each starting with a different system and/or task). After filling out a consent form, we follow the following procedure:

Part I: Introduction. (5 Minutes). We first explain the study procedure through an introductory video that explains the overall project scope by describing the problem setting (i.e., extracting claims from large document corpora). The video further shows the basic functionality of our tooling.

Part II: Solving Tasks. (60 Minutes). We let the users get set up with the respective tooling, and let them solve a specific task, each taking 15 minutes, followed by a standardized questionnaire.

(1) Get the tooling set up. As the baseline is the user's own preferred set of tools, the participants are prompted to open their note-taking application (e.g., Google Docs) and their software for organizing literature (e.g., Zotero). Additionally, depending on the user group they are asked to access the respective document search engines which are *PubMed* filtered on publicly accessible papers for the medical domain and a self-implemented keyword-based search on the UN speech corpus for the political one. The respective instantiation of our proposed workflow (i.e., the NEEDLE interface) for the political and medical domain was deployed, and could, therefore, be accessed through the web.
(2) We guide participants through the task by presenting the task description on a *Miro* collaborative sketch board. We briefly introduce the research topic (e.g., the effects of various diets on anxiety, Alzheimer's, and cancer) and ask participants to extract claims about these effects directly from scientific papers. To ensure traceability, we encourage them to take notes, cite references, and use direct quotes.
(3) We then let participants solve each task for 15 minutes, encouraging them to think aloud throughout. We only intervened if the user explicitly asked for help.
(4) After each task, we let users fill out a standard questionnaire about the system utility and perceived task load.

The user tasks are described in detail below:

- *Medical*: Various diets have been proposed to treat a range of different health problems. You are asked to gain an overview of how the ketogenic diet, intermittent fasting, and the Mediterranean diet affect/don't affect a set of health issues.
  - Task 1: Effects on anxiety, Alzheimer's, and cancer
  - Task 2: Effects on depression, Parkinson's, and cardiovascular disease.
- *Political*: You are investigating how different countries were speaking about specific conflicts between the years 2008 and 2022. You want to gather 1 to 2 statements that exemplify the use of the language of associated countries and their close allies on the topic.
  - Task 1: Russia - Ukraine conflict.
  - Task 2: Israel - Palestine conflict.

Part III: Unstructured Feedback. (10 Minutes). After both tasks had been completed, we let the users comment on the differences between the two workflows in an unstructured discussion.

Part IV: Feature Discussion. (10 Minutes). Finally, we let the user comment on each component of the workflow through a semi-structured approach, where we walk through each feature in a *Miro* collaborative sketch board.